TRUNKBELAY · PRECIS

TrunkBelay design precis

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-25	Amish Chadha	Kit 1.9.1 upgrade: the design is option A of TKB-OPT-001 (selection awaiting Amish, O5); fixed-end tether now a grade 80 connecting link through a 10 mm tether hole (TKB-DEC-001 D-S1); cost USD 828; requirement IDs PR-001 to PR-018 (TKB-PRS-001)
0.4	2026-10-03	Amish Chadha	Amish's decisions carried out (TKB-DDR-003); aluminium frame, tag-line haul, pre-rigged victim set; figures from TKB-CAL-001 v0.2
0.3	2026-10-03	Amish Chadha	TRL 3; constructable design of TKB-DDR-002 and figures from TKB-CAL-001
0.2	2026-10-03	Amish Chadha	TRL 2 precis; how it works, components with BOM numbers, key design choices, first-order numbers, safety
0.1	2026-09-30	Amish Chadha	Initial scaffold

TrunkBelay design precis

Lets a fellow climber anchor and lower a climber stranded high on a palm trunk.

This precis describes option A, the self-locking lever collar, which is the current, constructable design. Two other ways to anchor on the trunk (a constricting wrap anchor and a screwed stem anchor) are recorded in TKB-OPT-001; the choice between A and B is awaiting Amish (TKB-DEC-001 O5).

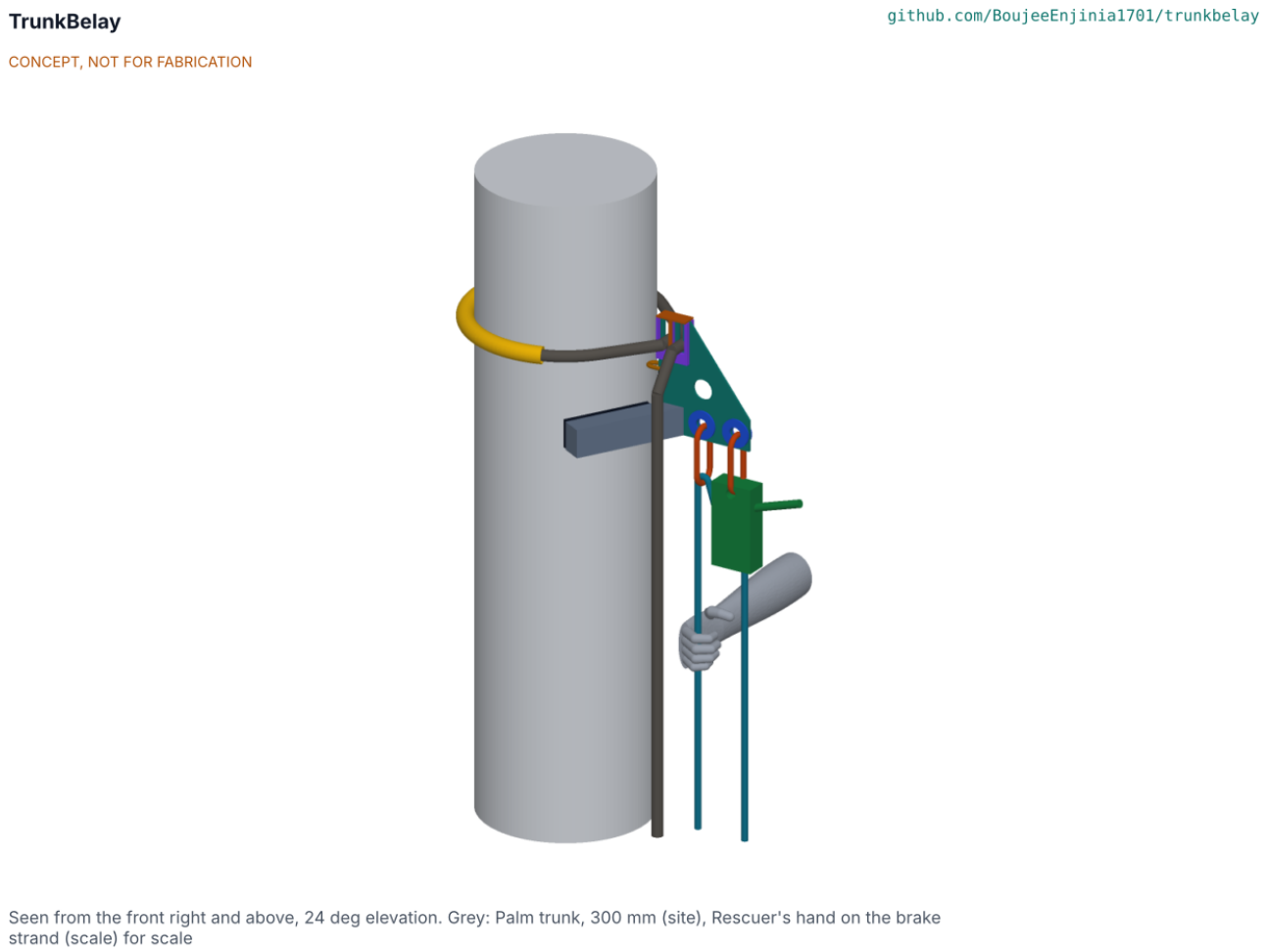


Figure 1. The collar on a 300 mm trunk, seen from the front right and above. The grey hand on the brake strand gives the scale.

How it works

TrunkBelay is an aluminium collar that grips a palm trunk harder the more load hangs from it, with a commodity rope descender clipped to it. The rescuer climbs on their own climbing device to just below the stranded climber, carrying the collar, the chain and the descender (about 4.7 kg) and a micro pulley with a 4 mm tag line running through it to the helpers on the ground, and works in four stages.

1. **Fix the collar.** The rescuer holds the collar frame against the trunk, with its two rubber-faced bearing bars touching the bark, passes the chain round the trunk, drops a link into the open slot on top of the frame, pulls the

chain snug and slips the keeper and lock pin on. The frame is set at least 300 mm above the point where the rope will meet the victim.

2. **Rig the lowering.** The helpers haul the rope bag up on the tag line. The descender is clipped to the load eye at the outer end of the frame and the rope reeved through it. The brake strand turns over a carabiner on the spare eye and runs down to the rope bag, which is lowered to the helpers on the ground.
3. **Take the weight.** The victim set comes out of the bag pre-rigged: the rope's end loop, the triangle's two waist loops and one end of the chest sling are already on the victim carabiner. The rescuer fits the triangle round the victim's hips and the sling round the chest, makes the last two clips (the crotch loop and the sling's free end) and pulls all slack through the descender, which locks.
4. **Free and lower.** The rescuer frees the victim's feet from the climbing device. The victim turns head up in the triangle and hangs from the collar. The rescuer opens the descender with its handle and lowers the victim to the helpers, holding the brake strand. The rescuer then takes the kit down and descends on their own device.

The load eye sits 150 mm out from the frame. The pull of the rope on the eye tilts the frame on the trunk: the bearing bars press the bark low on the near side and the chain presses it high on the far side. The friction at those two places rises in step with the load, so the collar is self-locking, like a drawer that jams when it is pulled from one corner (TKB-CAL-001, section B).

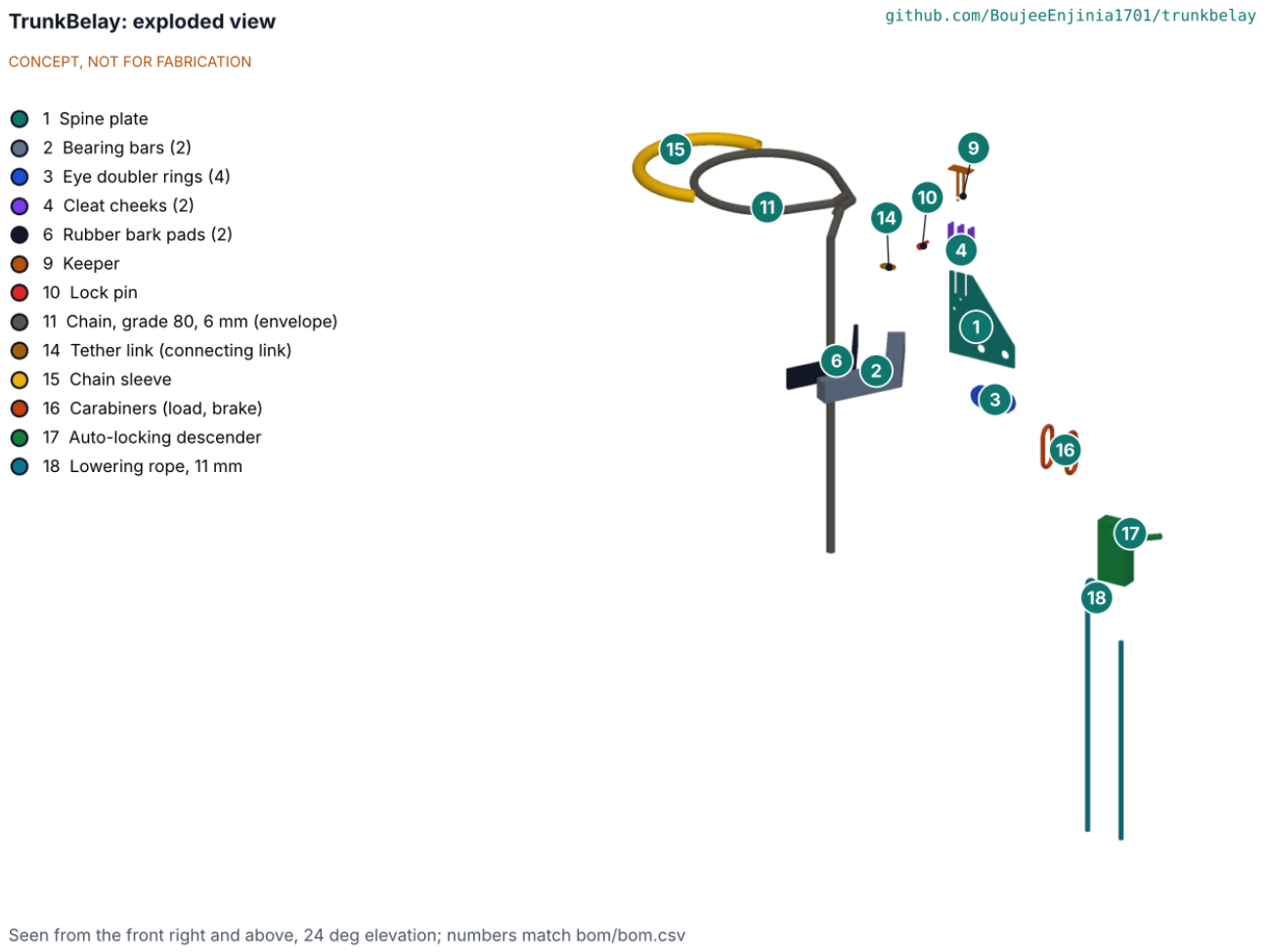


Figure 2. Exploded view; numbers match bom/bom.csv.

Components

Table 1. Components and their BOM lines.

BOM	Component	Role
1	Spine plate	5 mm 6082-T6 aluminium plate in the frame's centre plane: carries the bearing bars at its foot, the chain slots at its top, the load and spare eyes at its outer end and a 10 mm tether hole at its front edge
2	Bearing bars (2)	50 x 40 x 3 mm aluminium box section set in a 120 deg V at the foot of the spine; they bear on the bark through the pads
3	Eye doubler rings (4)	Thicken both eyes to 13 mm so a carabiner bears on a rounded, wide edge
4	Cleat cheeks (2)	8 mm aluminium; stiffen the slotted top of the spine; notched so the chain links bear on the 5 mm web
5	TIG welding	The frame is one TIG-welded aluminium piece, left bare
6, 7, 8	Rubber bark pads (2), screws, adhesive	10 mm rubber on each bar spreads the load on the bark and adds grip
9, 10	Keeper and lock pin	Hold both chain links down in their slots
11 to 14	Chain set	6 mm grade 80 chain, 110 links, with a master link joined by a connecting link on the adjustable end, and a second grade 80 connecting link tethering the fixed end through the tether hole (TKB-DEC-001 D-S1)
15	Chain sleeve	600 mm of tubular webbing on the chain where it bears on the back of the trunk
16	Carabiners (3)	Load eye to descender, brake redirect, victim connector
17	Auto-locking descender	Controls the lowering; locks if the rescuer lets go
18	Lowering rope	30 m of 11 mm semi-static rope
19, 20	Evacuation triangle and chest sling	Hold the victim head up once the feet are freed
21	Rope and carry bag	Holds the rope and the pre-rigged victim set; hauled up on the tag line, then lowered to the ground with the spare rope
22, 23	Tag line and micro pulley	55 m of 4 mm cord doubled through a micro pulley on the rescuer's harness; the helpers haul the bag up on it

Key design choices

All were made under Amish's pre-approval of 2026-10-03 and are recorded in TKB-DDR-001 and TKB-DDR-002. Amish decided the four requirement questions on 2026-10-03 (TKB-DDR-003): "i agree with all the 46 recommendations you provided. please proceed."

- **Self-locking frame, not a tightened band.** The grip comes from the load's lever on the frame, so it does not depend on how hard the rescuer pulled the chain. The chain only has to be snug.

- **A V of two bars, not a curved saddle.** A 120 deg V touches any trunk from 200 to 450 mm on two lines and centres itself; a curved saddle fits only one size.
- **A generic slotted cleat, no ratchet.** Chain links drop into two plain slots in the spine top and are held by a keeper and pin. This keeps clear of the ratcheting cinch geometry of the 3M DBI-SALA Cynch-Lok (the patent design-around) and has no moving parts to clog with bark.
- **Auto-locking descender.** The conservative choice: if the rescuer is hit, faints or lets go, the descender holds the victim.
- **Brake strand to the ground.** The spare rope goes down to the helpers, so the rescuer does not manage a rope bag at height.
- **Aluminium, TIG welded (decision 18C).** 6082-T6 plate and box section, sized on the weaker metal beside the welds; it takes about 1.9 kg off the climb compared with the same parts in steel. It needs an AC TIG welder rather than a stick welder, and nothing is galvanised.
- **The bag comes up on a tag line (decision 18C).** The rescuer carries only what fixes the collar and controls the lowering; the helpers haul the rope and the victim set.
- **Victim set packed pre-rigged (decision 17B).** Saves about 45 s at height, which offsets the time the haul takes.
- **The collar always goes above the victim's attachment (decision 16A).** The collar holds only a downward pull, so the drill keeps it at least 300 mm above the point where the rope meets the victim; a double-acting collar is the recorded fallback.
- **The safe rescue gear stays (decision 19A).** The auto-locking descender and the evacuation triangle are kept; cost is to come down through group purchase, not by cheaper gear.

First-order numbers

From TKB-CAL-001; assumptions are stated there.

Table 2. Key figures (estimates).

Quantity	Value
Trunk range	200 to 450 mm across, no tools
Working load on the collar, 100 kg person	0.99 kN
Static load the collar is sized for (R1)	2.5 kN
Friction needed to lock, 200 mm trunk (worst)	0.58, against about 0.9 assumed on wet bark; locking factor 1.54
Frame settles under 2.5 kN	About 10 mm
Strength at 2.5 kN	Frame 1.21 times its welded proof strength or better, even with all of the chain's pull on one slot (TKB-DEC-001 D-S2); chain 4.1 times its breaking force in one leg; rope 5.7 times at the knot; estimated breaking load about 5.0 kN, at or below the low end of rescue practice (TKB-DEC-001 O7)
Hand force on the brake strand, 100 kg	About 80 N
Rope	30 m; 28 m needed for a 25 m lowering
Rigging time once in position	About 4.5 minutes (estimate), the haul included
Mass	4.67 kg carried up the trunk; 3.7 kg bag hauled up; 8.4 kg in all
Cost	Value-engineering target: USD 900. Estimated cost of the constructable design: USD 828 (USD 72 under the target); PR-009 (R9 as restated), USD 850

Patent design-arounds

From the preliminary patent, trademark and prior-art screen (not legal advice):

- Generic chain-and-cleat collar; avoid the 3M Cynch-Lok ratcheting cinch geometry and cleated-plate arrangement. The constructable design has no ratchet, no cam and no cleated gripping plate: plain slots, rubber-faced bars and a lever.
- CA2681870C (Bashlin) lapsed in 2015; WO2012131439A1 ceased; neither blocks the concept.

Shared blocks

- CalRig proof-load (collar slip and proof tests on trunk sections)
- PalmSpan collar data (shelved sibling), for grip across palm sizes

Safety

Safety: TrunkBelay is rope rescue and work-at-height equipment. A slip of the collar, a wrongly fitted triangle or a lost brake hand can drop a person from 6 to 25 m. It is published as an open engineering reference, not certified rescue or fall-protection equipment.

- Only people trained in the drill use it, and they practise at low height first.
- Always call emergency services as well; TrunkBelay is for the minutes before they arrive.
- The collar is not used on a person before it has been proof-tested at 2.5 kN on cut trunk sections and the drill has been run with a dummy at low height (TRL 4). The build plan's safety stops gate both.
- The collar is loaded gently. All slack is taken in before the victim's feet are freed, so the victim never drops onto the collar (TKB-CAL-001, A2).
- The collar is set above the point where the rope meets the victim, so it is never pulled upward.
- A person hanging head down for a long time needs medical assessment after rescue.

This design is published as an open engineering reference. It is not certified equipment.